

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: North India Competitive Exam Aspirants (Hindi-Medium)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Individual MMLU items are culturally anchored in US/Western contexts: US Congress and constitutional history [\[DATASET-D1, D2, D3\]](#), US bar-exam tort scenarios [\[DATASET-D10, D11\]](#), Western philosophical canon [\[DATASET-D16, D17\]](#), US Social Security and birth-control demographics [\[DATASET-D20, D21\]](#), US measurement units in mathematics [\[DATASET-D25, D28\]](#), and CPA/MCAT institutional referents [\[DATASET-D32, D48\]](#). The benchmark assumes models acquired requisite knowledge from 'vast quantities of diverse Internet text' [\[Q60\]](#), which is overwhelmingly English-Western. The deployment requires both pan-India and North India-specific factual content (Chhath Puja, Panchayati Raj, tehsils, NCERT-aligned narratives) per elicitation Q2, none of which appears in the sampled items. Where India is referenced, it is via Pew Research statistics [\[DATASET-D23, D24\]](#) or colonial-British framing of the 1857 uprising [\[DATASET-D15\]](#) — both incompatible with NCERT/Bipin Chandra-aligned exam preparation.

ID	Evidence
Q60	"A major distinction between our benchmark and previous multitask NLP benchmarks is that we do not require large training sets. Instead, we assume that models have acquired the requisite knowledge from reading vast quantities of diverse text from the Internet. For this reason we assess pretrained models in a zero-shot, few-shot, or transfer setting and we provide a dev, val, and test set for each task." (p.7)
Q2	"To attain high accuracy on this test, models must possess extensive world knowledge and problem solving ability." (p.1)
D1	The right of American citizens to petition the government for a redress of grievances is protected under the First Amendment — US constitutional law question — irrelevant to Indian Polity
D10	A truck driver was assigned to drive a huge tractor-trailer loaded with logs ... the girl's parents ... assert a claim against the lumber company — US tort law scenario — calibrated against US bar exam, not Indian law
D16	According to Parfit, the obligation to give priority to the welfare of one's children is: agent-relative. — Western analytic philosophy (Parfit) — not covered in UPSC/SSC philosophy or ethics sections
D20	The age at which one can receive full benefits from social security ... Has been increasing — US Social Security system — irrelevant to Indian context
D25	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? ... 120 miles — Uses miles (US customary unit) — Indian exams use km; creates construct-irrelevant variance
D32	Which of the following procedures does a CPA usually perform when reviewing the financial statements of a nonissuer? — US CPA (Certified Public Accountant) professional standards — not relevant to Indian accounting exams
D23	As of 2017, what fraction of the population in India used the internet in the past three months? ... 26% — One of very few India-referencing questions — but tests dated Western-sourced statistical trivia, not Indian GK curriculum
D15	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing
D2	Both the War Powers Act of 1974 and the Budget and Impoundment Control Act of 1974 represent efforts by Congress to limit the powers of the president — US Congress-specific legislation — not Indian constitutional content
D3	What power was granted to the states by the Articles of Confederation but not by the Constitution? — US Articles of Confederation — no Indian Polity equivalent
D11	An avid baseball fan ... forwarded the club a \$50 registration deposit ... In a restitutionary action, can the executor of the fan's estate ... recover ... the \$2,500 paid to the club? — US contract law with baseball cultural context — no Indian law equivalent
D17	Anscombe claims that on Sidgwick's view, the badness of an action must be estimated in light of: its expected consequences. — Western analytic moral philosophy (Sidgwick, Anscombe) — entirely Western canonical tradition

D21	Which of the following is the most common method of birth control for married couples in the U.S. — Explicitly US-specific demographic question
D28	A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x? — Uses feet (US customary unit) — minor but consistent US measurement convention
D48	When preparing for the MCAT exam, a student begins studying electrochemical cells. — References MCAT (US Medical College Admission Test) — US-specific academic context
D24	As of 2018, about what percentage of Indians say that having an increasing number of people of many different races ... in their country makes it a better place to live ... 68% — India-referencing global survey question — but sourced from Pew/Western pollsters, not Indian exam curriculum

Strengths

- Pure-mathematics items (abstract algebra, probability) are content-culturally neutral and transfer cleanly [DATASET-D27, D30].
- Some basic NCERT-overlapping science facts (smallpox eradication, evolution) are factually correct in an Indian curriculum context [DATASET-D41, D42].

ID	Evidence
D27	Juan rolls a fair regular octahedral die ... What is the probability that the product of the two rolls is a multiple of 3? — Probability question — content universally applicable regardless of cultural context
D30	Find the characteristic of the ring $\mathbb{Z}_3 \times \mathbb{Z}_3$. — Abstract mathematics — entirely culture-independent
D41	The sequence of amino acids in hemoglobin molecules of humans is more similar to the hemoglobin of chimpanzees than it is to the hemoglobin of dogs. — Biology/evolution — curriculum-neutral STEM content
D42	Viruses have encouraged us to change our world, as we have now: Eradicated smallpox — General science fact — applicable to Indian science GK

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires India-centric cultural and administrative knowledge (festivals like Chhath Puja, tehsils, mandals, North Indian historical figures, NCERT-aligned narratives) per elicitation Q2; MMLU items do not address these.

IC-2: No — culturally sensitive content in MMLU (moral_scenarios explicitly framed as 'US 2020 moral standards' [DATASET-D18, D19]; jurisprudence using US product liability [DATASET-D33]) does not align with Indian deployment culture.

IC-3: Yes — extensive Western-specific knowledge requirements: US tax law [Q152], US bar-exam scenarios [Q153], US CPA standards [DATASET-D32], MCAT framing [DATASET-D48], US traffic conventions [DATASET-D45], and US customary measurement units in arithmetic [DATASET-D25, D28].

IC-4: INSUFFICIENT DOCUMENTATION — the paper does not document any review of items by Indian regional annotators; would need a systematic Indian SME audit to identify all culturally sensitive instances.

IC-5: Content validity is severely harmed: when India does appear [D15, D47], the framing privileges colonial historiography over the nationalist NCERT/Bipin Chandra narrative central to UPSC preparation, creating direct misalignment for exam-prep

use.

ID	Evidence
Q60	"A major distinction between our benchmark and previous multitask NLP benchmarks is that we do not require large training sets. Instead, we assume that models have acquired the requisite knowledge from reading vast quantities of diverse text from the Internet. For this reason we assess pretrained models in a zero-shot, few-shot, or transfer setting and we provide a dev, val, and test set for each task." (p.7)
Q2	"To attain high accuracy on this test, models must possess extensive world knowledge and problem solving ability." (p.1)
D18	"according to ordinary moral standards in the US as of 2020? Scenario 1 — The bomb shelter was filling up, but I made sure my grand piano got a spot."
D33	Bill purchased a can of Sipep from the Ajax Minimart ... In a strict product liability tort action against Ajax, Bill must prove ... — US-style product liability tort — uses American legal framework
D19	"according to ordinary moral standards in the US as of 2020? Scenario 1 — When I went through the road block I gave the policeman my brothers ID."
Q152	"Krete is an unmarried taxpayer with income exclusively from wages. By December 31, year 1, Krete's employer has withheld \$16,000 in federal income taxes and Krete has made no estimated tax payments. On April 15, year 2, Krete timely filed for an extension request to file her individual tax return, and paid \$300 of additional taxes. Krete's year 1 tax liability was \$16,500 when she timely filed her return on April 30, year 2, and paid the remaining tax liability balance. What amount would be subject to the penalty for underpayment of estimated taxes? (A) \$0 (B) \$500 (C) \$1,650 (D) \$16,500" (p.24)
Q153	"The night before his bar examination, the examinee's next-door neighbor was having a party. The music from the neighbor's home was so loud that the examinee couldn't fall asleep. The examinee called the neighbor and asked her to please keep the noise down. The neighbor then abruptly hung up. Angered, the examinee went into his closet and got a gun. He went outside and fired a bullet through the neighbor's living room window. Not intending to shoot anyone, the examinee fired his gun at such an angle that the bullet would hit the ceiling. He merely wanted to cause some damage to the neighbor's home to relieve his angry rage. The bullet, however, ricocheted off the ceiling and struck a partygoer in the back, killing him. The jurisdiction makes it a misdemeanor to discharge a firearm in public. The examinee will most likely be found guilty for which of the following crimes in connection to the death of the partygoer? (A) Murder. (B) Involuntary manslaughter. (C) Voluntary manslaughter. (D) Discharge of a firearm in public." (p.24)
D32	Which of the following procedures does a CPA usually perform when reviewing the financial statements of a nonissuer? — US CPA (Certified Public Accountant) professional standards — not relevant to Indian accounting exams
D48	When preparing for the MCAT exam, a student begins studying electrochemical cells. — References MCAT (US Medical College Admission Test) — US-specific academic context
D45	A flashing red traffic light signifies that a driver should do what? ... stop — US traffic law convention — minor but US-specific
D25	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? ... 120 miles — Uses miles (US customary unit) — Indian exams use km; creates construct-irrelevant variance
D28	A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x? — Uses feet (US customary unit) — minor but consistent US measurement convention
WEB-20	Translated MMLU variants suffer cultural mismatch because content (US History, Law) remains Western-specific even in Hindi (https://arxiv.org/abs/2509.11570)
D15	the peculiar aggravation of the Cawnpore massacres was this, that the deed was done by a subject race — by black men who dared to shed the blood of their masters ... William Howard Russell, My Indian Mutiny Diary, 1860 — 1857 Indian Mutiny from British colonial perspective — if Indian History content appears, it is filtered through colonial framing
D47	the deed was done by a subject race — by black men who dared to shed the blood of their masters ... British journalist William Howard Russell — 1857 mutiny through British colonial journalist's lens — if Indian content appears at all in MMLU, it is through a colonial/Western perspective

Information Gaps

- Full enumeration of India-relevant items beyond the ~200 sampled would require complete corpus review of all 14,079 test items.
- Annotator-side cultural assumptions are not documented in the paper.

Requires Expert Verification

- Indian SME review of moral_scenarios and jurisprudence items to determine extent of label flips under Indian normative frameworks.
- Audit of how 1857 / colonial history items would be scored against NCERT-aligned answer expectations.

Remediation

Gap 1: US measurement units (miles, feet) in mathematics; US institutional referents (CPA, MCAT, Social Security) introduce construct-irrelevant variance.

Recommendation: If MMLU mathematics items are reused, locally adapt unit conventions to SI/metric and replace US-institution-specific framings; alternatively, source quantitative-aptitude items directly from SSC CGL / UPSC CSAT past papers.